



LIBERTAS
JUSTITIA
VERITAS

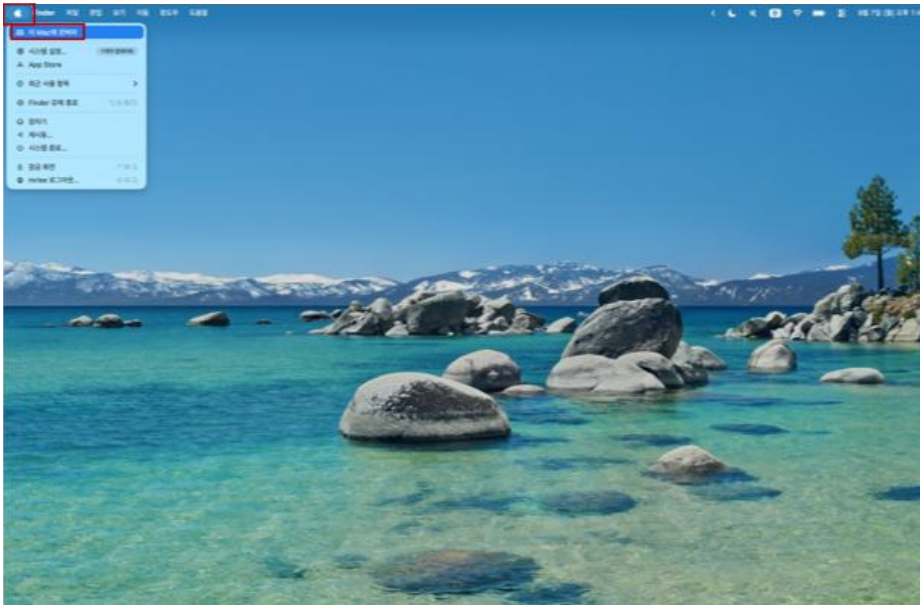
Fall 2026 Semester
Cryptographic Application

Setting for MacOS

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1. Check the Mac Processor

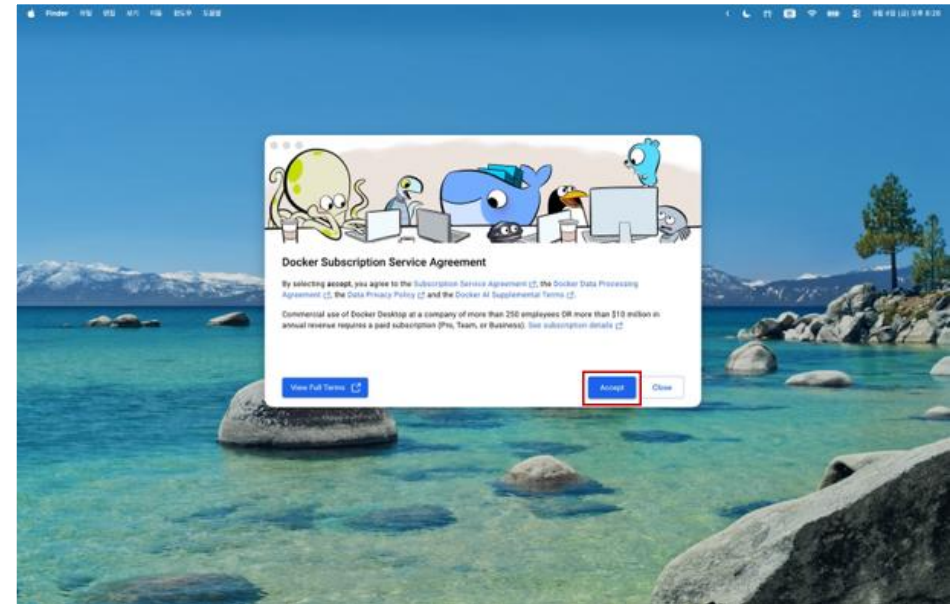
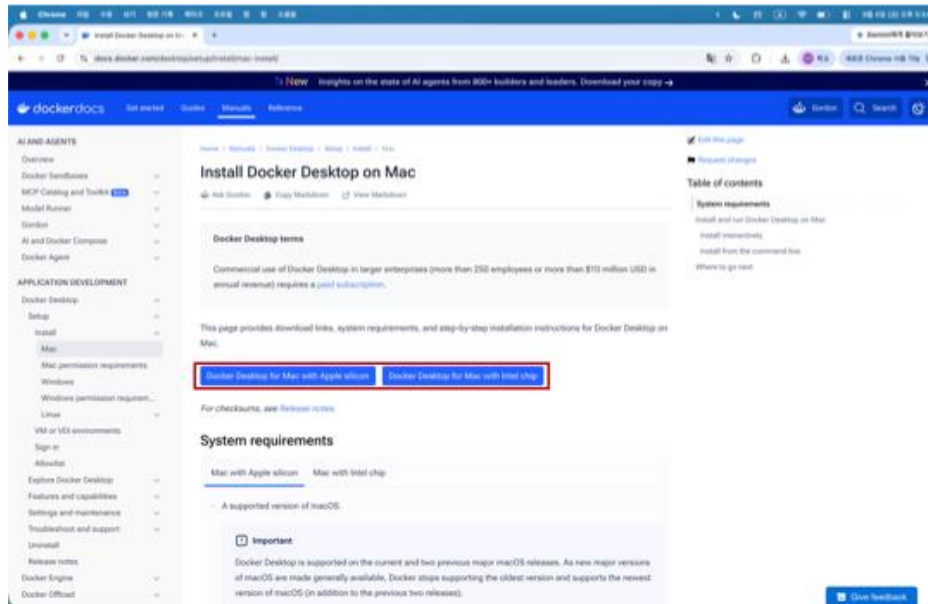
- Select Apple menu > About This Mac.
- Check whether the Mac uses an Apple chip or an Intel processor.



- If the Chip field shows an Apple M-series chip, your Mac uses **Apple silicon**.
- If the Processor field contains the word Intel, your Mac is an **Intel-based Mac**.

2. Install Docker Desktop

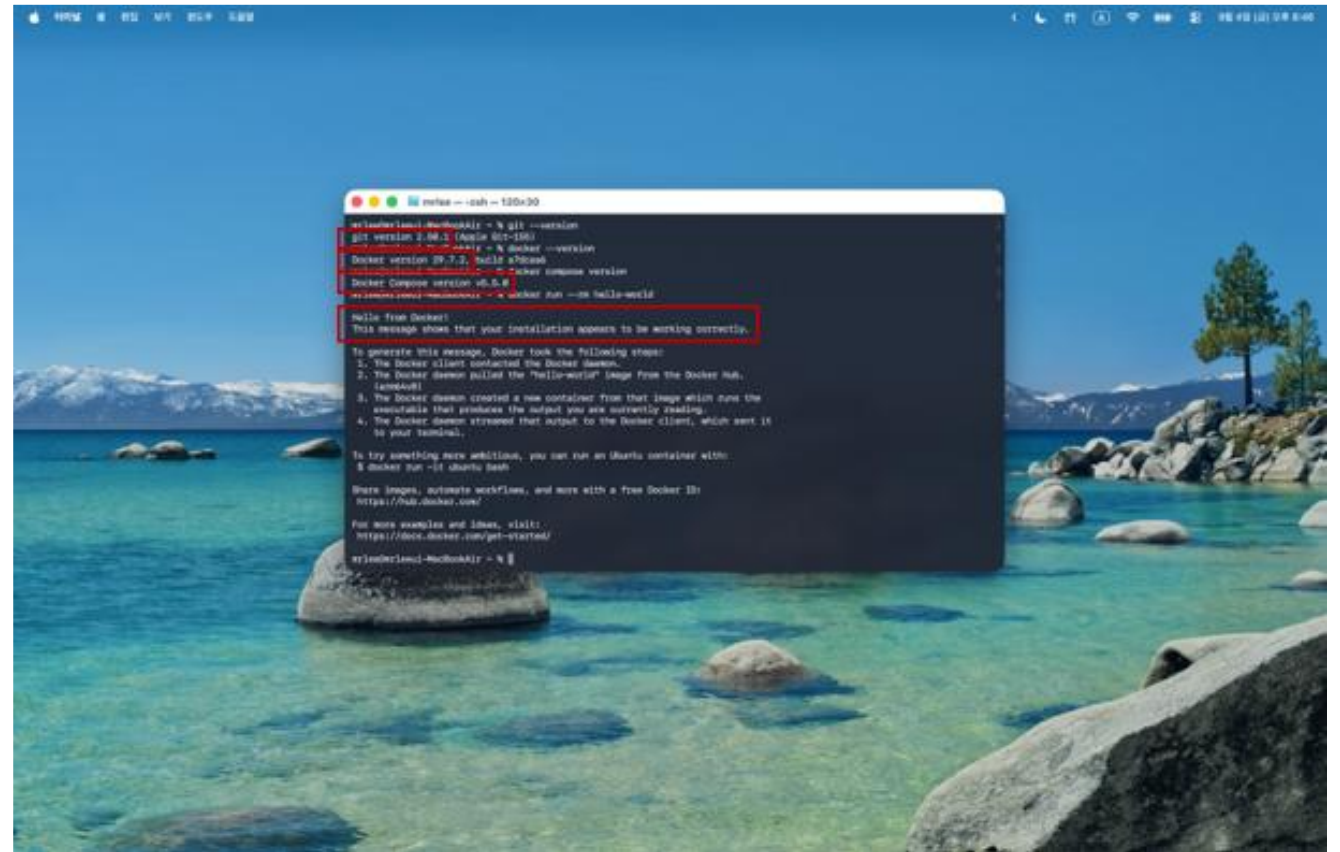
- Open the official Docker Desktop installation page for Mac.
→ <https://docs.docker.com/desktop/setup/install/mac-install/>
- Open Docker Desktop, accept the required terms, and use the recommended settings unless your institution instructs otherwise.



3. Verify the Required Tools

- Open Applications > Utilities > Terminal, then run:

```
git --version  
docker --version  
docker compose version  
docker run --rm hello-world
```



The screenshot shows a macOS desktop with a scenic background of a lake and mountains. A Terminal window is open, displaying the following commands and their outputs:

```
stefan@stefan-macbook: ~ % git --version  
git version 2.36.1 (Apple Git-136)  
stefan@stefan-macbook: ~ % docker --version  
Docker version 20.10.11, build e91ed17  
stefan@stefan-macbook: ~ % docker compose version  
Docker Compose version v2.0.0-rc.1  
stefan@stefan-macbook: ~ % docker run --rm hello-world  
Hello from Docker!  
This message shows that your installation appears to be working correctly.  
  
To generate this message, Docker took the following steps:  
1. The Docker client contacted the Docker daemon.  
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.  
   (amd64)  
3. The Docker daemon created a new container from that image which runs the  
   executable that produces the output you are currently reading.  
4. The Docker daemon streamed that output to the Docker client, which sent it  
   to your terminal.  
  
To try something more ambitious, you can run an Ubuntu container with:  
$ docker run -it ubuntu bash  
  
Share images, automate workflows, and more with a new Docker Hub:  
https://hub.docker.com/  
  
For more examples and ideas, visit:  
https://docs.docker.com/get-started/  
stefan@stefan-macbook: ~ %
```

4. Clone the Course Repository

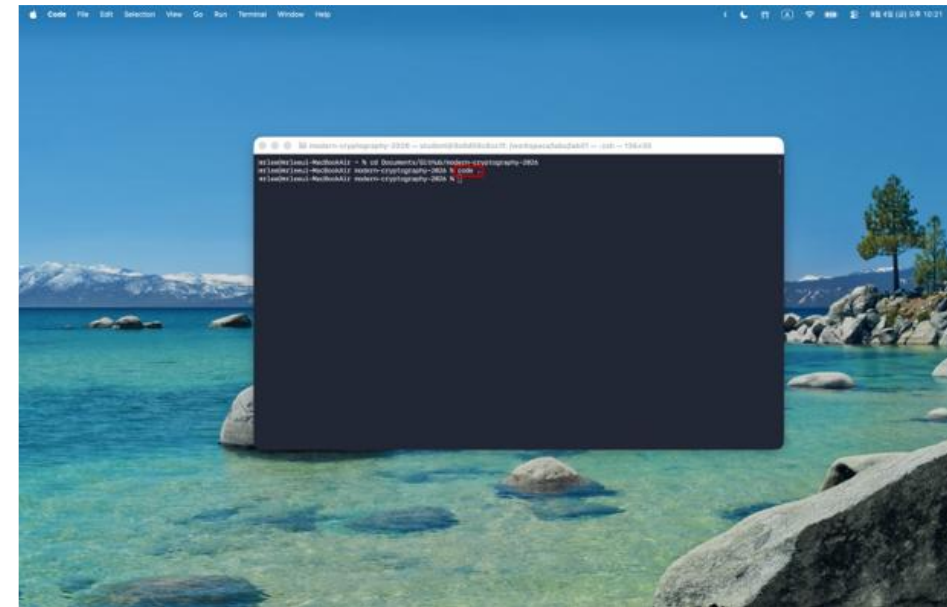
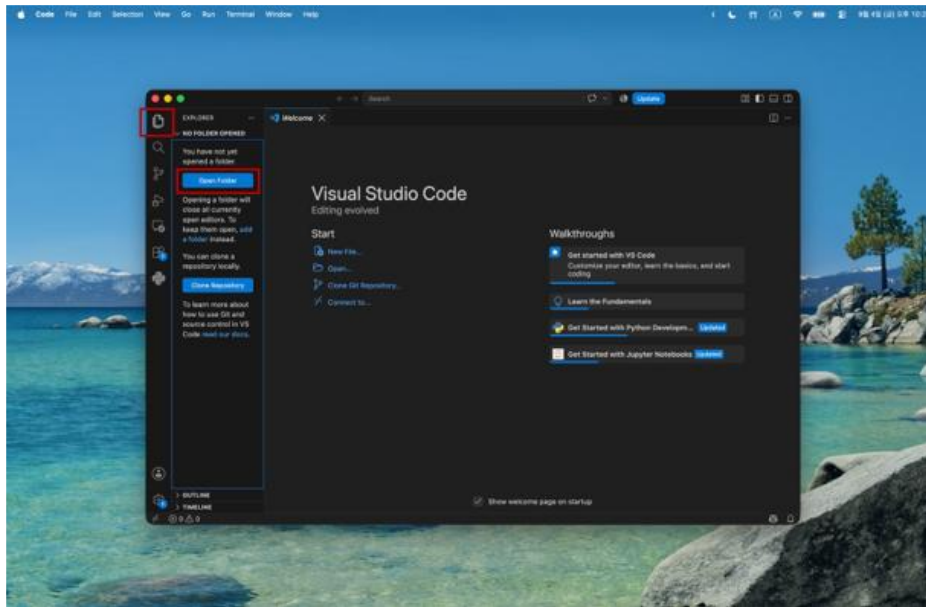
- **Run in: HOST - macOS Terminal**

```
git clone https://github.com/Wonsuk-EMSEC/modern-cryptography-2026.git  
cd modern-cryptography-2026
```



5. Open the Course Repository in VS Code

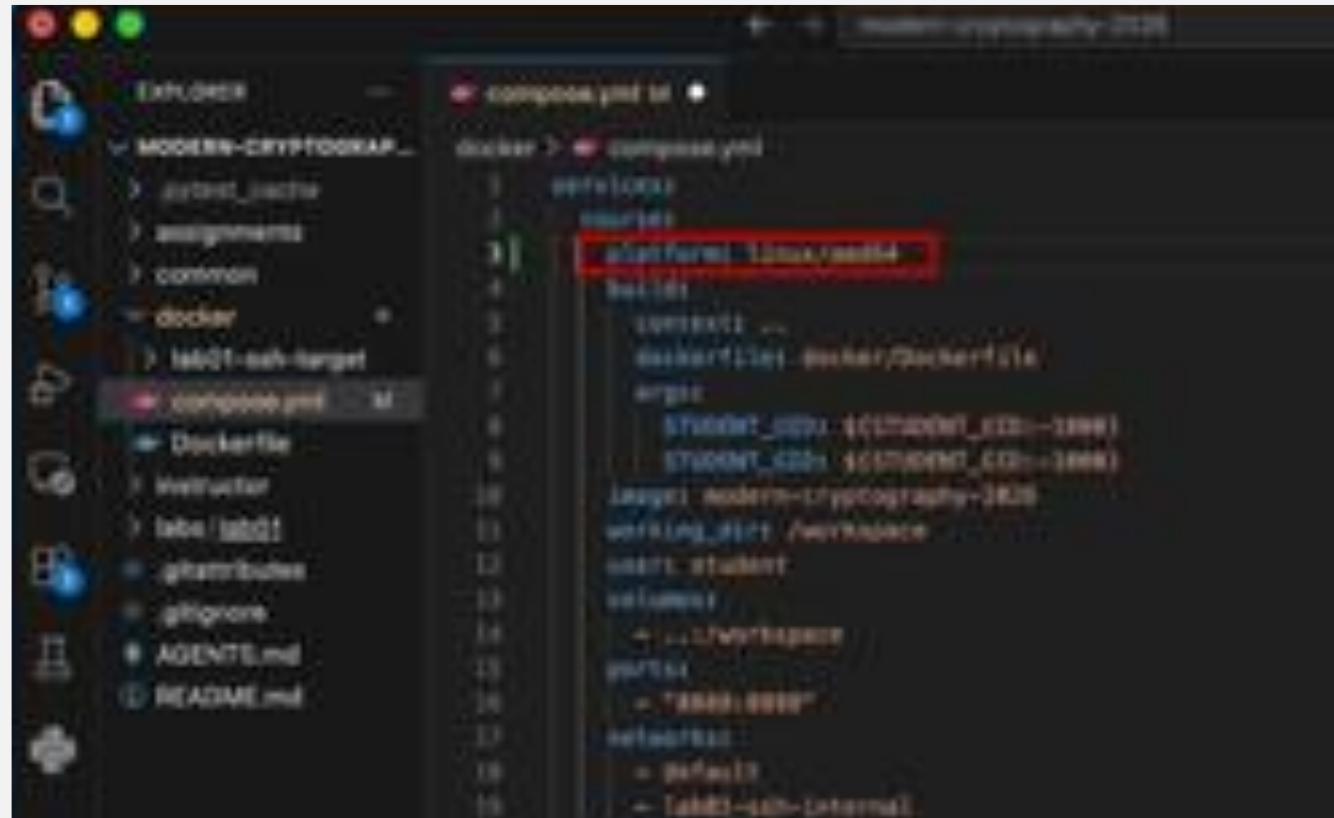
- Open Visual Studio Code and open the cloned modern-cryptography-2026 folder.
- If the code command is available, navigate to the cloned repository directory, then run: `Code .`



APPLE SILICON MACS ONLY

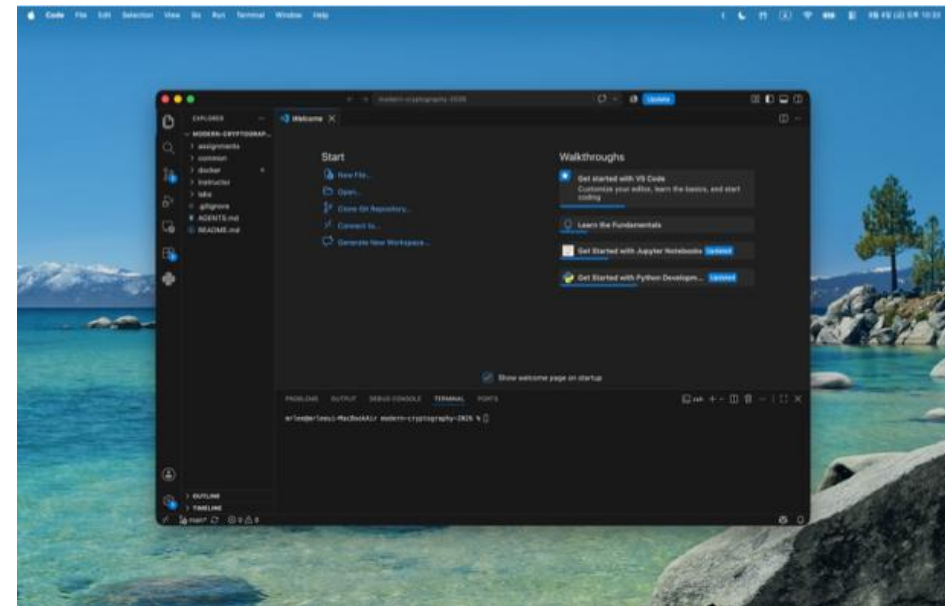
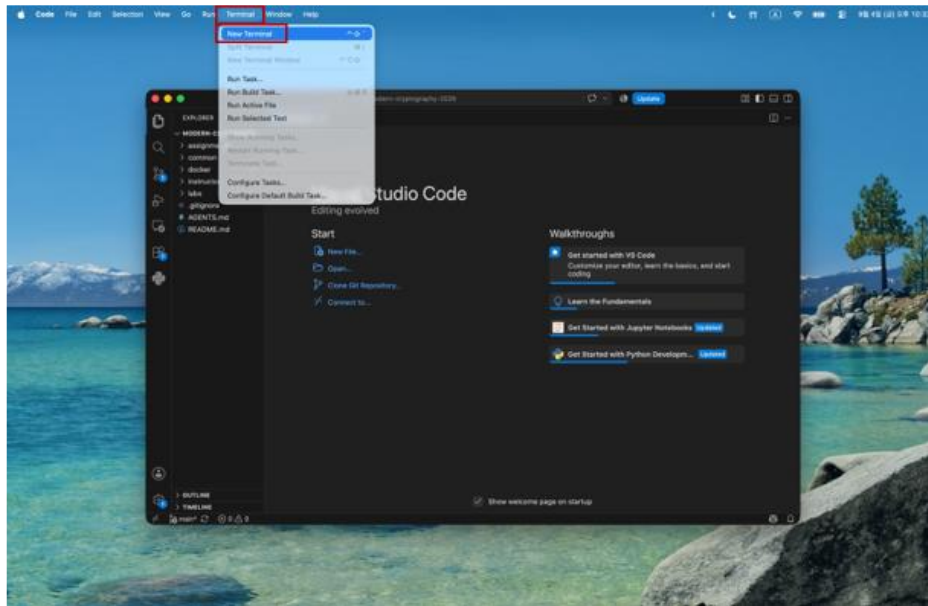
- Before building the course image, open docker/compose.yml and add:

```
platform: linux/amd64
```



6. Open the Visual Studio Code Terminal

- In Visual Studio Code, select Terminal > New Terminal. Confirm that the terminal is in the repository root.



7. Build and Start the Course Container

- The first time you use the course environment, build the Docker image:

```
docker compose -f docker/compose.yml build
```

- After the image has been built, start the course container:

```
docker compose -f docker/compose.yml run --rm course bash
```



8. Test the Lab Environment

- Run:

```
cd /workspace/labs/lab01  
pytest -q
```

